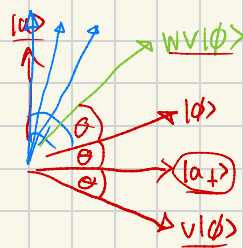


Assignment 10

Angle between $|\phi\rangle$ & $|a\rangle$

Angle between $|\phi\rangle$ & $|a\rangle_\perp$

$V \phi\rangle$	$\alpha + 2\theta$	$-\theta$
$WV \phi\rangle$	$\alpha - 2\theta$	$+3\theta$
$VRV \phi\rangle$	α	$+\theta$
$WRV \phi\rangle$	$\alpha - 4\theta$	$+5\theta$



2. On $|a\rangle$: $V|a\rangle = (I - 2|a\rangle\langle a|)|a\rangle$
 $= |a\rangle - 2|a\rangle\langle a|a\rangle = |a\rangle - 2|a\rangle\langle 1|$
 $= -|a\rangle$

So $|a\rangle$ get a phase -1 .

On $|b\rangle$ with $\langle a|b\rangle = 0$

$$V|b\rangle = (I - 2|a\rangle\langle a|)|b\rangle$$

$$= |b\rangle - 2|a\rangle\langle a|b\rangle = |b\rangle$$

So, vectors orthogonal to $|a\rangle$ are unchanged.

Compare with $W = 2|\phi\rangle\langle\phi| - I$

$$\text{on } |\phi\rangle : W|\phi\rangle = (2|\phi\rangle\langle\phi| - I)|\phi\rangle \\ = 2|\phi\rangle - |\phi\rangle = |\phi\rangle$$

so $|\phi\rangle$ is kept

on any $|\psi\rangle$ with $\langle\phi|\psi\rangle = 0$

$$W|\psi\rangle = (2|\phi\rangle\langle\phi| - I)|\psi\rangle \\ = 0 - |\psi\rangle = -|\psi\rangle$$

So everything orthogonal to $|\phi\rangle$ is flipped.

if $|\alpha\rangle = |\phi\rangle$ then

$$W = 2|\alpha\rangle\langle\alpha| - I = -I + 2|\alpha\rangle\langle\alpha| = -V$$

So they are the same reflection up to a global phase -1 .

$$3. |\phi\rangle = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} |x\rangle \\ = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} |x\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$|\alpha\rangle = |11\rangle$$

$$|\alpha\rangle = \frac{1}{\sqrt{2^n-1}} \sum_{x=0(x \neq n)}^{2^n-1} |x\rangle = \frac{1}{\sqrt{3}} \sum_{x=0(x \neq n)}^3 |x\rangle \\ = \frac{1}{\sqrt{3}} (|00\rangle + |01\rangle + |10\rangle)$$

$$4. V|\phi\rangle = V\left(\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)\right)$$

$$= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$5. \text{ Exponential speed up. } \frac{T_A}{T_B} = \frac{e^N}{2^N} = \left(\frac{e}{2}\right)^N \approx (1.3)^N$$

$$6. \langle 0| \langle 0| (U_f |0\rangle |0\rangle) = \langle 00|00\rangle = 1$$

$$\langle 0| \langle 1| (U_f |0\rangle |1\rangle) = \langle 01|01\rangle = -1$$

$$\langle 1| \langle 0| (U_f |1\rangle |0\rangle) = \langle 10|10\rangle = 1$$

$$\langle 1| \langle 1| (U_f |1\rangle |1\rangle) = \langle 11|11\rangle = 1$$

$$U_f = \langle 0| \langle 0| (U_f |0\rangle |0\rangle) |00\rangle \langle 00| + \\ \langle 0| \langle 1| (U_f |0\rangle |1\rangle) |01\rangle \langle 01| + \\ \langle 1| \langle 0| (U_f |1\rangle |0\rangle) |10\rangle \langle 10| + \\ \langle 1| \langle 1| (U_f |1\rangle |1\rangle) |11\rangle \langle 11| \\ = |00\rangle \langle 00| - |01\rangle \langle 01| + |10\rangle \langle 10| + |11\rangle \langle 11| \\ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = U_f$$

$$7. \text{ Firstly, } 2|01\rangle\langle 01| - I = 2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

We need to apply $H^{\otimes 2} (2|01\rangle_2 \langle 01|_2 - I) H^{\otimes 2} (V|\phi\rangle)$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

$$= -\frac{1}{4} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow 111 \text{ over rotated.}$$

$$8. V = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$V|\phi\rangle = V \left(\frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \right)$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\text{For } a=3 \quad H^{\otimes 2} (2|0\rangle_2 \langle 0|_2 - I) H^{\otimes 2} (V|\phi\rangle)$$

$$= \frac{1}{4} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle$$

$$9. C_Z = (I \otimes H) U_{\text{XOR}} (I \otimes H)$$

$$= \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \right)$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = C_Z.$$